School of Computer Science Engineering and Technology

Course-B.Tech	**Type-** Core
Course Code- CSET106	**Course Name-** Discrete Mathematical Structures
Year- 2024	**Semester-** Even
Date- 05/02/2026	**Batch-** 2025-2026

CO-Mapping

	Q1	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
CO1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CO2										
CO3										

Objectives: Students will be able to learn the proofs of classical set theory results.

Questions:

1. If E={4,8,12}and F={12,4,8}.Check whether sets E and F are equal.
2. G={2,4,6,8}and H={p,q,r,s}. Check whether sets G and H are equivalent.
3. Find the power set of following
   (a) A = {1}.
   (b) B = {a, b}.
   (c) C = {1, 2, 3}.
4. How many elements are there in the power set of a set containing 4 elements?
5. In a group of 60 people: 35 like Tea, 30 like Coffee, 15 like both Tea and Coffee
   Find:(a) Number of people who like only Tea
   (b) Only Coffee
   (c) At least one of the two
   (d) Neither Tea nor Coffee
6.
   a. In a survey of 200 students:120 like Discrete, 90 like Calculus, 70 like Java, 40 like both Discrete and Calculus, 30 like both Calculus and Java, 20 like both Discrete and Java. Everyone likes at least one of the three. How many people like exactly two subjects?

   b. How many integers from 1 to 500 are divisible by at least one of 4,6,10, but not divisible by all three?

7. If A = {1,2,3,4}, B = {1,3,5}. Then, compute
   a. $A \cap B$
   b. $A \cup B$
   c. A-B
   d. B-A
   e. $A \oplus B$
8. Define the partition of a set. Let X = {1,2,3,4,5,6,7}, then which of the following is the partition of X?
   a. S1={{1,2,3}, {3,4,5}, {6,7}}
   b. S2={{1,2}, {3,4}, {5,6,7}},
   c. S3={{1,2}, {3}, {5,6,7}}
9. In a competition, a school awarded 36 medals in dance, 12 in arts, and 18 in sports. Total 45 got the medals, and only 4 persons got all the three medals. How many students are there who got at least two medals?
10. In a competition, a school awarded 36 medals in dance, 12 in arts, and 18 in sports. Total 45 got the medals, and only 4 persons got all the three medals. How many students are there who got exactly two medals?